

14	B	$a_{\max} = \omega^2 x_o \text{ ----- (1) where } x_o = 0.30 \text{ cm}$ $\omega = \frac{2\pi}{T_{shm}} \text{ ----- (2)}$ $T_{shm} = \text{twice the period of } E_k - t \text{ graph [Remember this!]}$ $= 2 \times 0.20 \text{ s} = 0.4 \text{ s}$ $a_{\max} = \left(\frac{2\pi}{0.40}\right)^2 (0.30 \times 10^{-2}) = 0.74 \text{ m s}^{-2}$
15	A	With increased damping, the initial amplitudes are the same, the maximum amplitude